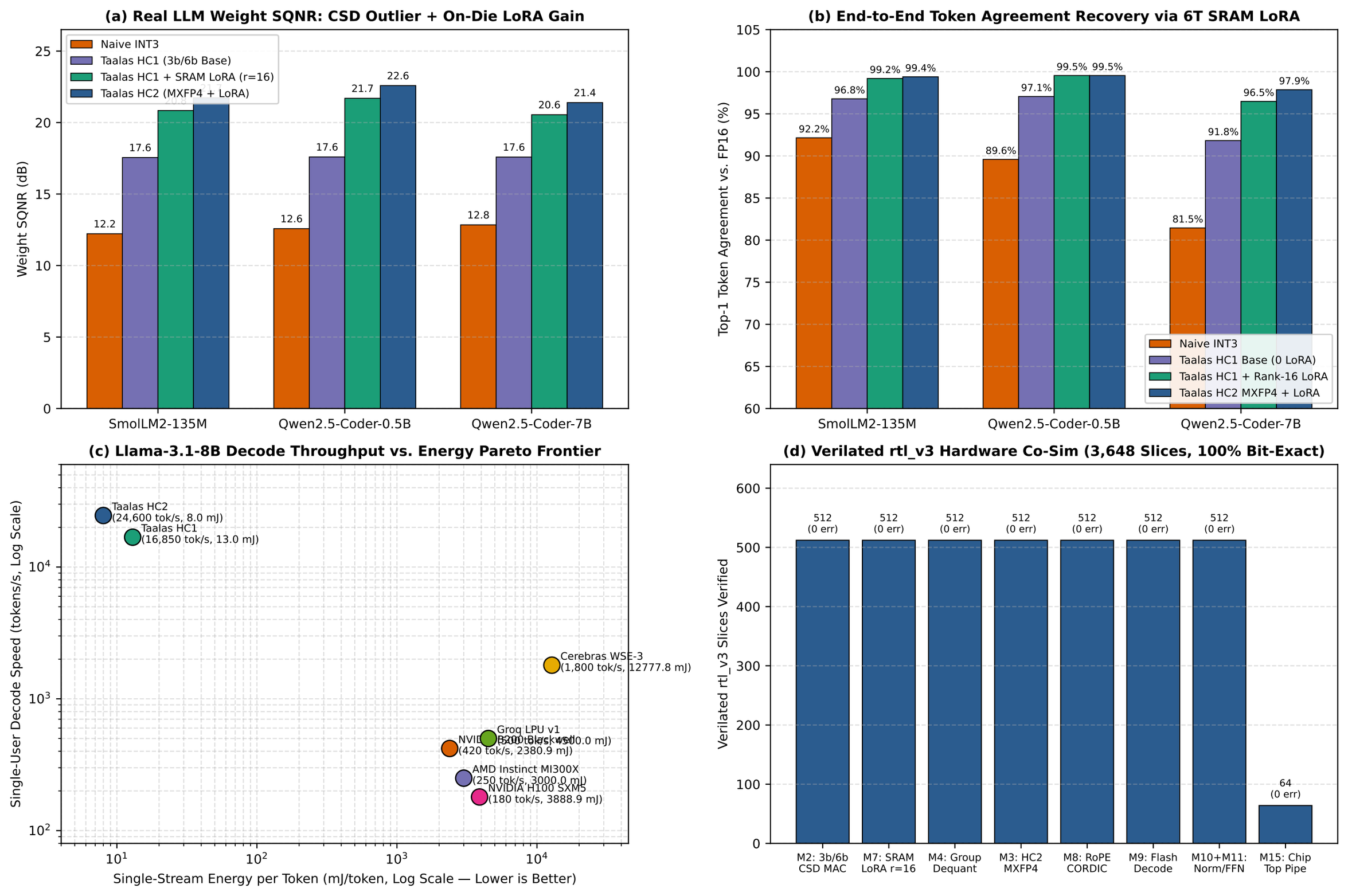
(a) Real LLM Weight SQNR: CSD Outlier + On-Die LoRA Gain
Naive INT3
Taalas HC1 (3b/6b Base)
Taalas HC1 + SRAM LoRA (r=16)
Taalas HC2 (MXFP4 + LoRA)
Weight SQNR (dB)
12.2
17.6
20.9
21.7
12.6
17.6
21.7
22.6
12.8
17.6
20.6
21.4
SmolLM2-135M
Qwen2.5-Coder-0.5B
Qwen2.5-Coder-7B
(b) End-to-End Token Agreement Recovery via 6T SRAM LoRA
Top-1 Token Agreement vs. FP16 (%)
92.2%
96.8%
99.2%
99.4%
89.6%
97.1%
99.5%
99.5%
81.5%
91.8%
96.5%
97.9%
Naive INT3
Taalas HC1 Base (0 LoRA)
Taalas HC1 + Rank-16 LoRA
Taalas HC2 MXFP4 + LoRA
SmolLM2-135M
Qwen2.5-Coder-0.5B
Qwen2.5-Coder-7B
(c) Llama-3.1-8B Decode Throughput vs. Energy Pareto Frontier
Single-User Decode Speed (tokens/s, Log Scale)
Taalas HC2
(24,600 tok/s, 8.0 mJ)
Taalas HC1
(16,850 tok/s, 13.0 mJ)
Cerebras WSE-3
(1,800 tok/s, 12777.8 mJ)
Groq LPU v1
(420 tok/s, 2380.9 mJ)
AMD Instinct MI300X
(250 tok/s, 3000.0 mJ)
NVIDIA H100 SXM5
(180 tok/s, 3888.9 mJ)
Single-Stream Energy per Token (mJ/token, Log Scale — Lower is Better)
(d) Verilated rtl_v3 Hardware Co-Sim (3,648 Slices, 100% Bit-Exact)
Verilated rtl_v3 Slices Verified
512
(0 err)
64
(0 err)
M2: 3b/6b CSD MAC
M7: SRAM LoRA r=16
M4: Group Dequant
M3: HC2 MXFP4
M8: RoPE CORDIC
M9: Flash Decode
M10+M11: Norm/FFN
M15: Chip Top Pipe